

PS6: Identifying structure in derivative problems

So exactly the hard part about learning derivative shortcut rules is figuring out which ones to use when. The direct answer to this question is, you use the shortcut rules that match the **algebraic structure** of the function whose derivative you are finding.

Identifying **algebraic structure** takes some practice! Let's get into it with this problem set.

You will need six different color pens, pencils, highlighters, or whatever, and I recommend printing this sheet out and writing on it directly.

Structure rules

- ☐ sum rule:

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

- ☐ constant multiple rule: if c is a constant, then

$$\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}[f(x)]$$

- ☐ product rule (aka rule for functions that are multiplied):

$$\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}[f(x)] \cdot g(x) + \frac{d}{dx}[g(x)] \cdot f(x)$$

- ☐ quotient rule (aka rule for functions that are divided):

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \cdot \frac{d}{dx}[f(x)] - f(x) \cdot \frac{d}{dx}[g(x)]}{[g(x)]^2}$$

Function rules

- ☐ constant function rule:

If c is a constant, then

$$\frac{d}{dx}[c] = 0$$

(Every constant function is flat lol)

- ☐ power function rule:

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

- ☐ exponential function rule:

$$\frac{d}{dx}[a^x] = a^x \cdot \ln(a)$$

trigonometric function rules:

☐ $\frac{d}{dx}[\sin(x)] = \cos(x)$

☐ $\frac{d}{dx}[\cos(x)] = -\sin(x)$

1. Consider the function

$$f(t) = \frac{7}{t} - 7^t + 3t^7 - 2 \sin(t) + \sqrt{t}$$

In different colors for each:

- (a) circle anywhere that functions are being *added or subtracted* to make a bigger function,
- (b) circle everything that's a *constant multiple* of some function,
- (c) circle everything that's a *power function* (hint: there are three!),
- (d) circle everything that's an *exponential function*,
- (e) and lastly circle everything that's a *trigonometric function*.

Finally, now that you have done all this circling, find $\frac{d}{dt}[f(t)]$ using the appropriate shortcut rules. The more work you can show, the more confidence you will gain in the process!

2. Consider the function*

$$g(x) = (2x^2 - 5x - 3) \cos(x)$$

In different colors for each:

- (a) circle anywhere that functions are being *multiplied* to make a bigger function,
- (b) circle anywhere that functions are being *added or subtracted* to make a bigger function,
- (c) circle everything that's a *constant multiple* of some function,
- (d) circle everything that's a *power function* (hint: there are two!),
- (e) circle everything that's a *constant function*,
- (f) and lastly circle everything that's a *trigonometric function*.

Finally, now that you have done all this circling, find $\frac{d}{dx}[g(x)]$ using the appropriate shortcut rules. The more work you can show, the more confidence you will gain in the process!

*I pulled this function from [CheckIt practice problems DF4!](#)

3. Consider the function[†]

$$h(y) = \frac{e^y}{3y^2 + 3y - 5}$$

In different colors for each:

- (a) circle anywhere that functions are being *divided* to make a bigger function,
- (b) circle anywhere that functions are being *added or subtracted* to make a bigger function,
- (c) circle everything that's a *constant multiple* of some function,
- (d) circle everything that's a *power function* (hint: there are two!),
- (e) circle everything that's a *constant function*,
- (f) and lastly circle everything that's an *exponential function*.

Finally, now that you have done all this circling, find $\frac{d}{dy}[h(y)]$ using the appropriate shortcut rules. The more work you can show, the more confidence you will gain in the process!

[†]This function is also from [CheckIt practice problems DF4!](#)